

thermal camera and spray nozzles
on the fixed skirt

exhaust leaves backward

arriving PuffSat gas,
 $w = 46\text{--}68\text{ km/s}$,
footprint $\approx$ half the radius

2 m skirt: barrier-coated maraging liner,
carbon hoop wrap (16 t)

argon spray cloud,
 $\sim 4\text{ m}$ deep, $\sim 1\text{ m}$ off the floor

skirt frame: the skirt is fixed to the vehicle and never moves with the floor

Vectran gas springs

vehicle

thrust

maraging floor, 98 t, tapered,
 $\sim 30\text{ mm}$ mean; floor and band wrap
(134 t) are the sprung mass

carbon hoop wrap on the steep band (36 t)

dashed: floor at full stroke

Face (thicknesses not to scale)

pitch film, 225–300 μm standing
alumina hydrogen barrier,
under 1 μm

ceramic felt

thermal camera:
hot spots mark thin film,
respray only there

→ gas springs

maraging 300, 2.5 GPa face limit

Rim joint (not to scale)

to the vehicle

skirt: 2 mm liner + carbon wrap

floor kick $\sim 100\text{ m/s}$

sprung, segmented seal;
gap leak vented outward,
away from the gas springs

The floor slides like a piston. A rigid joint
would carry a $\rho c \Delta v$ stress wave of 1–3 GPa
within the $\sim 1\text{ ms}$ pulse, so the skirt is not
attached to the floor.